

SHAKIR AHMED

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EDUCATION

Master of Science Physics University of Cologne	01 October 2022 – Present Cologne, Germany
Bachelor of Science Physics Shahjalal University of Science and Technology	01 April 2015 – 30 August 2019 Sylhet, Bangladesh
Higher Secondary Certificate Science Govt. Science College	03 August, 2013 Dhaka, Bangladesh

WORK EXPERIENCE

Working Student - Algorithms R&D Software, Luna Innovations Germany GmbH	01 April 2024 – 31 March 2026 Cologne, Germany
- Architected and deployed full data preparation, training, evaluation, and prediction pipelines for industrial deep learning models utilizing PyTorch, TensorFlow, and DVC for high-precision temperature and acoustic signal analysis. - Engineered data processing pipelines to automatically index, search, and process massive HDF5 datasets utilizing FastAPI, PostgreSQL, and Streamlit. - Engineered a scalable image annotation workflow by integrating Label Studio with AWS S3, while developing custom COCO compatibility for internal applications. - Authored comprehensive technical documentation and contributed to rigorous final project reports for stakeholders.	
Subject Matter Expert Academics Team, Shikho Technologies Bangladesh Ltd	05 September 2021 – 30 November 2022 Dhaka, Bangladesh
- Translated abstract technical physics concepts into structured, accessible presentation frameworks. - Developed 50+ comprehensive lesson scripts aligned with the national Physics and Mathematics curriculum. - Directed the design and animation pipeline to ensure high-quality, visually accurate educational content.	
System Administrator Hardware Team, HEXit	10 March 2017 – 15 September 2017 Sylhet, Bangladesh
- Administered and maintained Linux servers running Apache, MySQL, DNS, SSH, and FTP services. - Conducted continuous system performance monitoring. - Implemented network topologies and administered DNS infrastructure to ensure efficient system management. - Implemented cybersecurity measures to mitigate system vulnerabilities and prevent unauthorized intrusions.	

PROJECTS AND RESEARCH

Neural Operator Mapping for Acoustic Dynamics ↗ A personal project implementing Geometry-aware Fourier Neural Network	May 2026 – June 2026
Wigner Monte Carlo Approach to Neutrino Oscillation Python ↗ Shahjalal University of Science and Technology	August 2019 – December 2019
On the Relativistic Dynamics of Electromagnetic Two-Body Problem Python ↗ Shahjalal University of Science and Technology	September 2018 – February 2019
Radioactive Waste Disposal in Outer Space ↗ The 2016 University Physics Competition ↗	12 November 2016

🔧 TECHNICAL SKILLS & LANGUAGES

AI & Machine Learning / MLOps: Python, PyTorch, TensorFlow, Keras, Scikit-Learn, Pandas, h5py, DVC, MLflow, Prefect.
Scientific Computing: SciPy, NumPy, MATLAB, SageMath, C++, Qiskit, Signal Processing, Spectral Analysis.
DevOps: Git, CI/CD, Docker, Linux (Arch/Debian, Bash), Windows (PowerShell), Web (HTML5, CSS3, JS, Node.js).
Software & Communication: $\text{\LaTeX}$, Jira, MS Office (Word, PowerPoint, Excel), Adobe (Photoshop, Lightroom, Premiere Pro).
Languages: Bengali (Native), English (Fluent - C1), German (Beginner - A2).

SEMINARS AND WORKSHOPS

Fundamental Error Threshold in Quantum Codes [↗](#)

14 April – 30 June, 2025

Advanced seminar on Open Quantum Systems [↗](#), Institute for Theoretical Physics, University of Cologne

Cologne, Germany

8th Workshop on Collaborative Scientific Software Development and Management of Open Source Scientific Packages

19 October – 09 December, 2021

The Abdus Salam International Centre for Theoretical Physics

Trieste, Italy

HONORS AND AWARDS

Silver Medal

15 January 2017

The 2016 University Physics Competition [↗](#)

CONFERENCES AND PRESENTATIONS

Simulation of Neutrino Oscillation using Wigner Monte Carlo

05 - 07 March 2020

International Conference on Physics - 2020, Atomic Energy Centre

Dhaka, Bangladesh

Numerical Analysis of the Dynamics of Two Charged Particles System [↗](#)

07 - 09 February 2019

National Conference on Physics - 2019, University of Dhaka

Dhaka, Bangladesh

CERTIFICATIONS

Applied Data Science II: Machine Learning and Statistical Analysis (Honors) [↗](#)

06 December, 2021

WorldQuant University

Applied Data Science I: Scientific Computing and Python (Honors) [↗](#)

11 September, 2021

WorldQuant University

Advanced Machine Learning and Signal Processing [↗](#)

17 August, 2020

IBM

Deep Learning Specialization [↗](#)

11 June, 2020

DeepLearning.AI